

Week 8: Review of Weeks 1–7

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Session 8.3 Homework

Instructions: Show all work. Circle your final answers.

Topics Covered:

- Week 3: Inequalities (Compound & Word Problems)
- Week 4: Linear Equations (Forms & Intercepts)
- Week 5: Systems of Linear Equations (Graphing & Classification)
- Week 6: Linear Functions (Slope & Applications)
- Week 7: Systems of Inequalities & Applications

Quick Reference: Key Formulas & Concepts

Compound Inequalities

AND: $a < x < b$ (between two values)

OR: $x < a$ OR $x > b$ (outside two values)

Linear Equation Forms

Slope-intercept: $y = mx + b$

Point-slope: $y - y_1 = m(x - x_1)$

Standard form: $Ax + By = C$

Finding Intercepts

x-intercept: Set $y = 0$ and solve for x

y-intercept: Set $x = 0$ and solve for y

Slope Formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad m = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x}$$

Systems Classification

One solution: Different slopes

No solutions: Same slope, different y-intercepts (parallel)

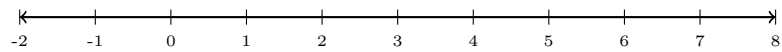
Infinitely many: Same slope, same y-intercept (same line)

Part A: Compound Inequalities

Problem 1

Solve: $5x - 3 \leq 7$ OR $2x + 1 > 9$

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Problem 2

A taxi charges a \$4 base fee plus \$2.50 per mile.

You have at most \$30 to spend.

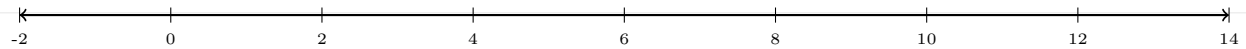
Let m = miles traveled (in miles).

Assume $m \geq 0$.

Write and solve an inequality for m .

State the greatest whole number of miles you can travel.

(a) Graph the solution on a number line.



(b) Write the solution in interval notation.

Part B: Linear Equations - Forms & Intercepts

Problem 3

Convert to standard form ($Ax + By = C$ where A, B, C are integers):

$$y = \frac{3}{4}x - 5$$

Problem 4

Find the x-intercept and y-intercept of: $5x - 2y = 20$

Problem 5

Write the equation of the line through $(2, -3)$ with slope $\frac{1}{2}$ in point-slope form.

Part C: Systems - Graphing & Classification

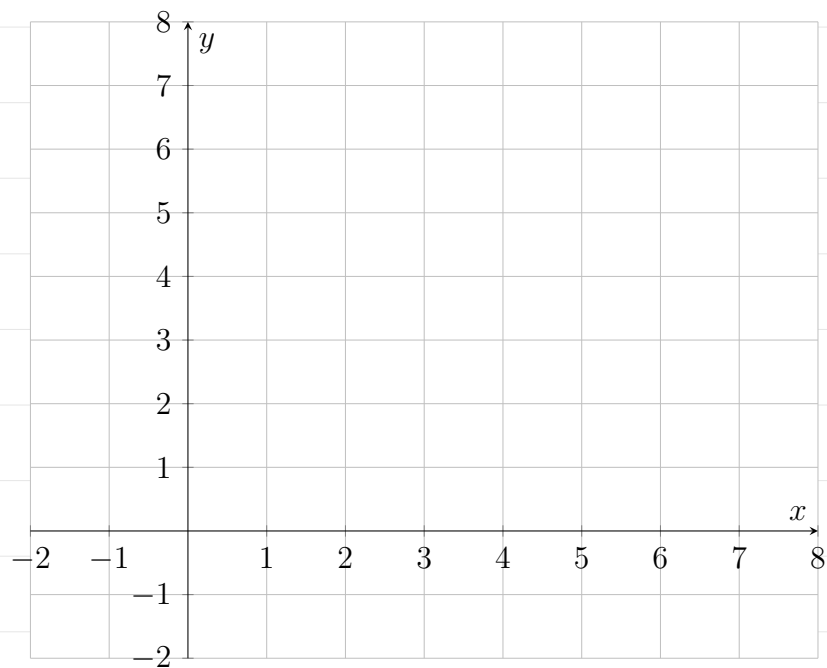
Problem 6

Graph each line and classify the system (one solution / no solutions / infinitely many solutions).

Find the solution if it exists.

$$y = \frac{1}{2}x + 1$$

$$y = -x + 4$$



Problem 7

Without graphing, classify the system:

$$2x + 3y = 6$$

$$4x + 6y = 18$$

Part D: Slope & Applications

Problem 8

A rental car company charges \$45 per day plus \$0.20 per mile.

Let C = total cost (in dollars) and m = miles driven.

(a) Write an equation for C in terms of m for one day.

(b) Find the cost for driving 150 miles in one day.

Part E: Systems of Inequalities

Problem 9

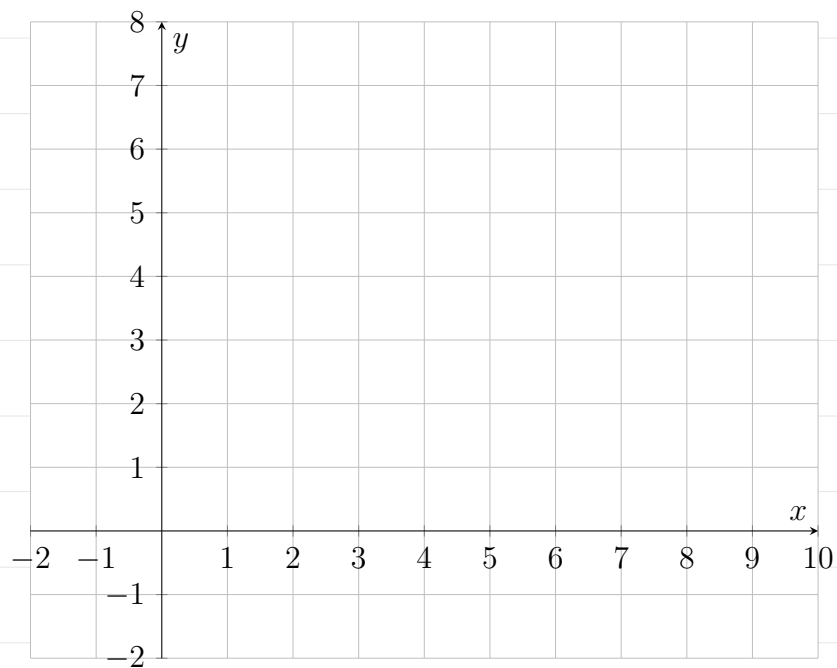
Graph the system of inequalities and shade the solution region:

$$y \geq x - 2$$

$$y < -\frac{1}{2}x + 4$$

$$x \geq 0$$

$$y \geq 0$$



Part F: Applications & Word Problems

Problem 10: Mixture Problem

A chemist needs to prepare 80 mL of a $15 \frac{mg}{mL}$ acid solution.

She has a $10 \frac{mg}{mL}$ acid solution and a $25 \frac{mg}{mL}$ acid solution.

How much of each should she mix?

Problem 11: Break-Even Problem

A small business makes custom t-shirts.

The fixed costs are \$500 per month.

Each t-shirt costs \$8 to produce.

Each t-shirt sells for \$20.

How many t-shirts must be sold to break even?

Problem 12: System Word Problem

Concert tickets are sold in two types: floor seats and balcony seats.

There were 200 tickets sold in total.

Floor seats cost \$75 each.

Balcony seats cost \$50 each.

The total revenue was \$12,500.

How many of each type of ticket was sold?